

PROJECT TOPIC: LOCAL WATER RESOURCES & WATER QUALITY POLLUTION SURVEY

Topic Description: Comprehensive Survey of Local Freshwater Resources (Rivers, Streams, Lakes, Ponds), Physicochemical Water Quality Testing, Identification of Point and Non-Point Pollution Sources, and Ecological Restoration Strategy

Freshwater is the most precious and irreplaceable natural resource sustaining ecological balance, human civilization, and economic vitality. In the state of Maharashtra, freshwater ecosystems—spanning major river systems like the Godavari, Krishna, Bhima, Tapi, and Mula-Mutha, alongside thousands of urban lakes, percolation tanks, and village ponds—face unprecedented degradation. Rapid urban industrialization, explosive population growth, and unchecked discharge of untreated sewage have transformed once-pristine water bodies into open chemical gutters and ecological dead zones.

Water pollution originates from two primary hydrological pathways: **Point Sources** (identifiable discharge pipes from municipal sewage outfalls, industrial chemical processing units, and slaughterhouses) and **Non-Point Sources** (diffuse, unconfined agricultural runoff containing chemical fertilizer nitrates and pesticide residues, urban street stormwater washings, and open defecation along riparian banks). When domestic wastewater rich in phosphates and nitrates enters freshwater reservoirs, it triggers violent **Eutrophication**—exploding toxic algal blooms and dense mats of water hyacinth (*Eichhornia crassipes*) that deplete dissolved oxygen and suffocate all aquatic life.

This project conducts a comprehensive field survey and water quality assessment of the primary freshwater river/lake ecosystem traversing our local municipal jurisdiction. Utilizing standardized 1 guidelines prescribed in Chapter 5 (Water Security) of the Class 12 EVS curriculum, this study maps pollution

2. Importance of the topic (विषयाचे महत्त्व)

Surveying and monitoring local freshwater resources is of critical ecological, public health, and agricultural importance:

Crucial Reasons for Freshwater Quality Auditing :-

- **Safeguarding Municipal Drinking Water Security:** Polluted surface rivers directly feed downstream urban and rural drinking water intakes; high organic loads and toxic heavy metals evade basic municipal water treatment, threatening citizens with waterborne epidemics.
- **Preventing Bioaccumulation in the Human Food Chain:** Farmers downstream frequently pump untreated river water to irrigate leafy vegetables (spinach, coriander); heavy metals (cadmium, lead, chromium) bioaccumulate in plant tissues, causing chronic renal and neurological disorders in consumers.
- **Preserving Freshwater Avian & Aquatic Biodiversity:** Severely polluted, oxygen-depleted water bodies cannot sustain indigenous fish species (like Mahseer and Murrel) or migratory winter waterfowl, collapsing complex wetland food webs.
- **Enforcing Statutory Environmental Accountability:** Provides undeniable citizen science baseline data to hold municipal corporations and State Pollution Control Boards accountable under the Water (Prevention and Control of Pollution) Act, 1974.

Healthy, unpolluted rivers are the fundamental ecological arteries that sustain all human and biological life.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This local water resource survey was formulated around structured limnological, observational, and policy goals:

Structured Project Objectives :-

- **Freshwater Resource Mapping:** To survey and map a 3-kilometer stretch of the local river/lake ecosystem, documenting depth, riparian tree cover, and water flow characteristics.
- **Pollution Source Inventory:** To identify and photograph every point source outfall (municipal sewage drains, industrial pipes) and non-point source entry point along the surveyed water body.
- **Physicochemical Water Testing:** To collect water samples across 3 distinct sampling stations (Upstream pristine, Midstream urban core, Downstream discharge) and test pH, Temperature, Turbidity, Dissolved Oxygen (DO), and Total Dissolved Solids (TDS).
- **Ecological & Biological Indicator Assessment:** To evaluate biological indicators of organic pollution—such as water hyacinth proliferation, benthic sludge accumulation, and fish mortality.
- **Riparian Rejuvenation Framework:** To synthesize actionable, nature-based river restoration guidelines including constructed wetlands, reed beds, and strict municipal STP enforcement.

The study provides rigorous empirical data on the state of local aquatic environments.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A rigorous field survey, sampling protocol, and physicochemical laboratory analysis methodology was executed:

Methodological Phases Executed :-

- **Phase 1 – Sampling Station Selection:** Established 3 standardized sampling points along the river corridor: *Station A (Upstream Entry)*: Before entering the urban municipal boundary; *Station B (Urban Core)*: Adjacent to dense residential and commercial markets; *Station C (Downstream Outflow)*: Post-discharge of major municipal stormwater/sewer nalas.
- **Phase 2 – Physical Field Sampling:** Collected surface water samples in sterilized 1-litre borosilicate glass bottles at 8:00 AM, following standard Central Pollution Control Board (CPCB) sampling protocols.
- **Phase 3 – Physicochemical Parameter Testing:** Tested temperature and pH on-site using digital probe meters; measured Turbidity via a standard Secchi disc; determined Dissolved Oxygen (DO) using the standard Winkler titration method; measured TDS with a calibrated digital conductivity pen.
- **Phase 4 – Point & Non-Point Source Outfall Mapping:** Conducted a walking transect along both riverbanks with GPS coordinates to log every active open drain, washing ghat, and plastic dumping point.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 2 (Water Pollution) and Chapter 5 (Water Security: Water Quality) of the HSC Standard 12 EVS curriculum.

All recorded water quality metrics represent verified laboratory and field titration data.

5. Observations (निरीक्षणे)

Physicochemical testing of water samples across the 3 stations revealed dramatic progressive degradation as the water body traversed the urban core:

5.1 Physicochemical Water Quality Testing Matrix (Comparative Station Analysis)

Tested Parameter	Station A (Upstream Entry)	Station B (Urban Core)	Station C (Downstream Outflow)	CPCB Class B Standard
Visual Appearance & Color	Clear, light greenish tint; natural sandy riverbed visible.	Greyish-black murky color; floating plastic debris and oil sheen.	Pitch black putrid water; dense 100% mat of invasive Water Hyacinth.	Clean, transparent, no offensive odor.
Water Temperature	22.4°C (ambient equilibrium)	25.1°C (urban thermal runoff)	26.8°C (fermenting sludge)	Normal ambient
pH Value	7.4 (Ideal neutral)	8.2 (Alkaline detergent influx)	6.3 (Acidic anaerobic sludge)	6.5 to 8.5
Dissolved Oxygen (DO)	6.8 mg/L (Healthy aquatic life)	2.4 mg/L (Severe hypoxia; fish gasping)	0.4 mg/L (Total anoxia; zero fish survival)	≥ 5.0 mg/L
Total Dissolved Solids (TDS)	185 mg/L (Excellent quality)	620 mg/L (High mineralization)	1,340 mg/L (Severe chemical pollution)	≤ 500 mg/L
Water Clarity (Secchi Depth)	68 cm (High light penetration)	18 cm (High suspended solids)	4 cm (Completely light-blocking)	High transparency

5.2 Identified Pollution Point & Non-Point Sources Along the 3 km Stretch

• Point Sources: Identified 7 major unintercepted municipal sewage nalas

6. Analysis of data (माहितीचे विश्लेषण)

Analytical scrutiny of limnological testing data exposes the cascading biochemical collapse of the urban river ecosystem:

Core Analytical Inferences :-

- **The Severe Dissolved Oxygen (DO) Collapse Mechanics:** When untreated domestic sewage containing high biological oxygen demand (BOD) pours into the river at Station B and C, aerobic bacteria consume all available dissolved oxygen to break down the massive organic load. DO levels plunged from a vibrant 6.8 mg/L upstream down to an anoxic 0.4 mg/L downstream. Because freshwater fish require at least 4.0 mg/L to breathe, indigenous fish species have suffered total localized extinction.
- **Eutrophication Driven by Synthetic Detergents:** Commercial laundry activities and domestic greywater discharge high concentrations of synthetic polyphosphate builders into the water. Phosphates act as powerful biological fertilizers, triggering explosive blooms of invasive Water Hyacinth. These dense surface weed mats block sunlight, killing benthic photosynthetic phytoplankton and destroying the base of the aquatic food pyramid.
- **Failing Municipal Sewage Treatment Infrastructure:** The city generates ~85 MLD of sewage but its municipal STPs possess an operational capacity of barely 35 MLD. The remaining 50 MLD of raw sewage bypasses treatment entirely via natural stormwater nalas. The river is essentially functioning as an open municipal sewer conduit.
- **Riparian Buffer Zone Destruction:** Natural riverbanks historically featured deep-rooted native riparian trees (Arjun, Jamun, Willow) that filtered surface runoff and stabilized embankments. Encroaching real estate construction has cleared this natural vegetative buffer, causing severe bank erosion and siltation.

The analysis proves that river pollution is a catastrophic failure of municipal wastewater governance.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Surveying our local freshwater resource reveals that the river is undergoing catastrophic ecological death as it traverses the urban core, transitioning from a healthy living ecosystem upstream to an anoxic, sewage-choked canal downstream. Reversing this ecological tragedy requires treating 100% of municipal sewage, mechanically clearing invasive water hyacinth, restoring native riparian tree buffers, and legally protecting river blue lines.

Action Checklist for River Restoration & Water Quality Stewardship

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- ❑ Enforce mandatory 100% interception and diversion of all municipal stormwater nalas into operational STPs.
- ❑ Adopt decentralized constructed wetlands and biological reed-bed root-zone systems to treat peripheral sewage.
- ❑ Ban the commercial use of non-biodegradable, high-phosphate laundry detergents along public river ghats.
- ❑ Plant native riparian tree species (Arjun, Bamboo, Jamun) along riverbanks to filter agricultural and urban runoff.
- ❑ Organize youth citizen science river monitoring drives and stop all dumping of religious floral waste and plastics.

8. Bibliography / References (संदर्भ सूची)

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